Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-02-01 To:2026-02-31

Project No: IT6010 Student No: MS25905922

Project Title: Research Project [2026/JAN]

Name of the student: U. D. K De Silva

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

Topic confirmed

Section B - Work carried out during the last month:

By analyzing current research papers to establish a focus and identify gaps in the existing literature.

Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? Yes/~~No~~

Comments (please comment on problems, recommendations, improvements, etc.):

Name of the supervisor: ~~Prof~~./Dr/~~Mr/Ms~~ Dinuka R. Wijendra

Signature:…… Date: 2026/08/15

Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-03-01 To:2026-03-31

Project No: IT6010 Student No: MS25905922

Project Title: Research Project [2026/JAN]

Name of the student: U.D.K De Silva

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

Research proposal document made

Section B - Work carried out during the last month:

Through research conducted using research papers identify proper gaps with the help of the supervisor and made the research proposal. Approved by the supervisor and the panel.

Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? Yes/~~No~~

Comments (please comment on problems, recommendations, improvements, etc.):

Name of the supervisor: ~~Prof.~~/Dr/~~M~~/~~M~~s Dinuka R. Wijendra

Signature:…… Date: 2026/08/15

Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-04-01 To:2026-04-30

Project No: IT6010 Student No: MS25905922

Project Title: Research Project [2026/JAN]

Name of the student: S.A.A Perera

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

The architectural blueprinting is complete. The baseline system is operational, establishing the control metrics (unoptimized latency and costs) required for future benchmarking.

Section B - Work carried out during the last month:

Finalized the overarching research scope by mapping out the latency and API cost bottlenecks associated with traditional cloud-based RAG architectures.

Developed and engineered "System A" (the Naive RAG pipeline) to serve as the project's baseline control model.

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Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? Yes/~~No~~

Comments (please comment on problems, recommendations, improvements, etc.):

Dinuka R. Wijendra

Name of the supervisor: ~~Prof./~~Dr/~~Mr/Ms~~…….

Signature:…… Date: 2026/08/15 ……..

Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-05-01 To:2026-05-31

Project No: IT6010 Student No: MS25905854

Project Title: Research Project [2026/JAN]

Name of the student: S.A.A Perera

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

The data engineering and vectorization phase is complete. Both the foundational itinerary knowledge base and the slang glossary are sanitized, mathematically encoded, and ready for database seeding.

Section B - Work carried out during the last month:

Curated and processed the primary dataset of authentic Sri Lankan destination reviews, enforcing strict quality control and tagging entries with a specific `@district`taxonomy.

- Manually created the structured "Local Tourism Glossary" to capture hyper-local terms (eg, *tuk-tuk* , *kottu*).
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- Built Python ingestion scripts and successfully encoded both datasets into vectors using the`instructor-base` embedding model.

Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? ~~Yes~~/No

Comments (please comment on problems, recommendations, improvements, etc.):

Name of the supervisor ~~Prof~~./Dr/~~Mr/Ms~~…Dinuka R. Wijendra…….

Signature:…… Date: 2026/08/15

Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-06-01 To:2026-06-30

Project No: IT6010 Student No: MS25905854

Project Title: Research Project [2026/JAN]

Name of the student: S.A.A Perera

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

The core microservice infrastructure is officially online. The system's "Fast Path" is functional, allowing the architecture to mathematically intercept repetitive travel queries.

Section B - Work carried out during the last month:

Deployed the localized, cloud-independent Docker environment, successfully networking the FastAPI backend, Redis Stack, PostgreSQL (`pgvector`), and the local 4-bit quantized Llama-3 model.

Designed and integrated the core Redis Semantic Caching layer into the orchestrator, establishing a strict `> 0.85`cosine similarity threshold for query interception.

Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? Yes/~~No~~

Comments (please comment on problems, recommendations, improvements, etc.):

Name of the supervisor: ~~Prof.~~/Dr/~~Mr/Ms~~ Dinuka R. Wijendra

Signature:…… Date: 2026/08/15

Sri Lanka Institute of Information Technology
M. Sc. Research Project
Monthly Progress Report

From:2026-07-01 To:2026-07-31

Project No: IT6010 Student No: MS25905854

Project Title: Research Project [2026/JAN]

Name of the student: S.A.A Perera

Instructions:

1. The objective of this progress report is to help students to complete their research project in time and to help the supervisors to track the progress of their students.
2. This report contains 3 sections. The student must submit this report to the supervisor after completing Sections A and B. The supervisor will complete the Section C and send the report to the Coordinator /M.Sc. Research Projects.
3. Submitting monthly progress reports in a timely manner is a responsibility of the student. Failing to submit monthly progress reports may affect the final grade obtained for the project.

Section A - Current status of the project:

The defensive logic and cultural translation integrations are fully complete. The System B architecture is fully functional, secured against semantic collisions, and ready for high-concurrency JMeter load-testing in Month 5.

Section B - Work carried out during the last month:

Developed and deployed the system's custom hallucination guards to protect the cache, specifically the Geographic Pre-Filtering taxonomy (`@district`) and the Hybrid Lexical Jaccard Penalty Trap.

Wired the vectorized PostgreSQL glossary directly into the query routing path to act as a "2-Way Cultural Translation Bridge," injecting accurate local slang definitions into the LLM prom

Section C – (To be completed by the supervisor)

Instructions: Please complete the following section and return the report to the Coordinator /M.Sc. Research Projects directly.

(Tick the appropriate cell)

	Good	Satisfactory	Needs improvement	Not satisfactory
Current status of the project		✓		
Work carried out during the last month		✓		
Timely completion of work according to the initial proposal		✓		
Maintaining regular contacts with the supervisor		✓		

Are you satisfied with the present overall progress of the research? Yes/~~No~~

Do you see any serious problems in the current work that may affect the timely completion of the project? Yes/~~No~~

Comments (please comment on problems, recommendations, improvements, etc.):

Need to finalize the jounral paper with first few sections.

Name of the supervisor: ~~Prof./~~Dr/~~Mr/Ms~~ Dinuka R. Wijendra

Signature:…… Date: 2026/08/15